12th hacker series

Ecosystem

Types of ecosystem

Broadly divided into two basic categories

- **Terrestrial**
  - eg - Forest
  - desert
- **Aquatic**
  - eg - river
  - ocean

Similarly, with respect to human interference, the ecosystem are of two types

1) natural ecosystem- river, Ocean ,Forest , desert
2) anthropogenic ecosystem or man-made ecosystem- aquarium, agro ecosystem

Characteristic of anthropogenic ecosystem

- Do not possess self regulatory mechanism
- Have little diversity
- Simple food chain
- High productivity
- Little cycling of nutrients



Ecosystem - structure

- important structural features include:
- species composition
  - identification, and enumeration of plant and animal species of an ecosystem
  - Tropical rainforest is dense with amazing number of biological species
  - Vegetation is sparse in the desert ecosystem
- stratification
  - vertical distribution of different species occupying different levels
  - For example, in a forest following vertical subdivision are present
  - a) top layer- trees
  - b) second layer- Shrubs
  - c) bottom layer- grasses, and herbs

Ecosystem function

- the key functional aspect of the ecosystem are:
- 1) productivity
- 2) decomposition
- 3) energy flow
- 4) nutrient cycling

Productivity

- rate of biomass production
- Two types:
- 1) primary- formed at producer level
- 2) secondary- formed at consumer level

Primary productivity

- rate at which biomass or organic matter is produced by plants or producers during photosynthesis per unit area
- Expressed in terms of weight as $(g\,m^{-2})\,yr^{-1}$
- Energy as $(Kcal\,m^{-2})\,yr^{-1}$

- divided into two categories:
- 1) gross primary productivity (GPP): rate of production or synthesis of organic matter by producers during photosynthesis per unit time and area
- 2) net primary productivity (NPP): rate of organic matter, buildup or stored by producers in excess of respiratory utilisation, per unit time and area

$$NPP = GPP - R$$

- note: net primary productivity is the available biomass for the consumption to heterotrophs i.e, both herbivores and decomposers

Factors affecting primary productivity

- photosynthetic capacity of producer
- Solar radiations available
- Temperature
- Soil moisture
- Availability of nutrients

Productivity of biosphere

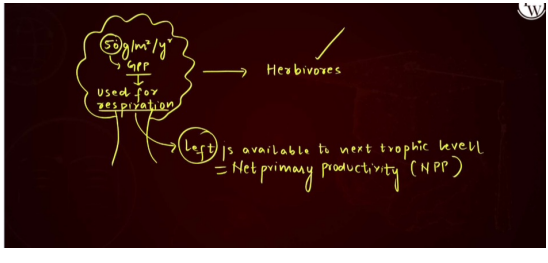
- 170 billion tonnes (dry weight) of organic matter
- The productivity of oceans are only 55 billion tons
- For terrestrial ecosystem is 115 billion tons

Reasons for low productivity of oceans

- light- it decreases with depth
- Nutrients- most limiting nutrient of marine ecosystem is nitrogen

Secondary productivity

- rate of formation of new organic matter by consumers



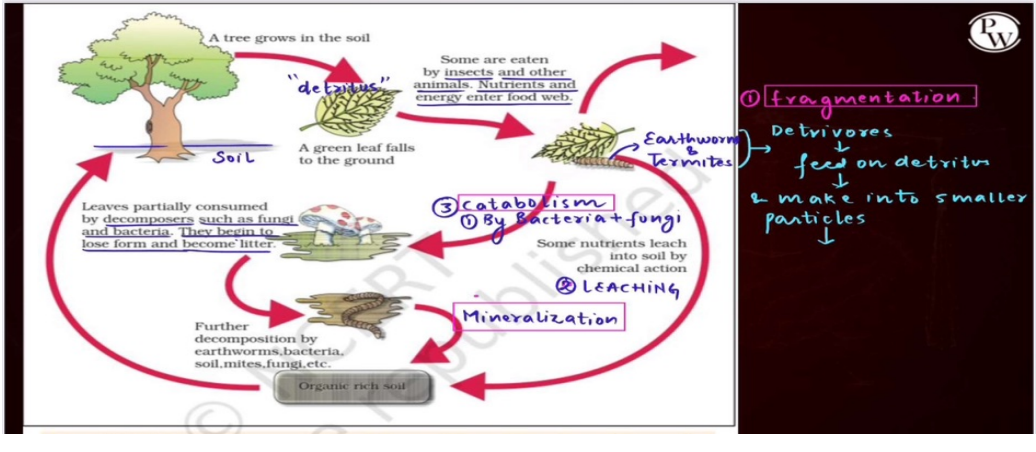
Decomposition

- dead remains of plants and animals

Factors affecting decomposition

- chemical nature of detritus- if it is rich in tannins, decomposition is slow; if it is rich in water soluble substance, decomposition is fast
- Temperature- less than 10°C, slow decomposition
- Moisture
- aeration

- temperature and soil Moisture are the most important climatic factors to regulate decomposition through their effects on the activities of soil microbes
- Warm and moist environment, favour decomposition
- Low temperature and anaerobiosis inhibit decomposition, resulting in buildup of organic material



Energy flow

- unidirectional
- Sun ⟶ Producers ⟶ Consumers

Energy flow follows laws of thermodynamics

- first law of thermodynamics- energy is neither created nor destroyed, but can be transformed from one state to another
- Second law of thermodynamics- no transfer of energy occurs, unless and until it is accompanied by degradation or dissipation of energy

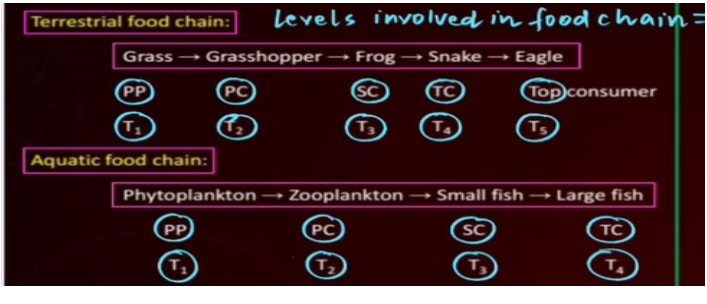
Food chain

types of food chain

- grazing food chain or predatory food chain
- Detritus food chain or saprophytic food chain
- Parasitic food chain or auxiliary food chain

Grazing food chain

- consists of:
- 1) producers
- 2) consumers
- 3) decomposers
- Source of energy for such food chain is sun
- it is major conduit of energy flow in aquatic ecosystem



Detritus food chain

- begins with detritus or dead organic matter
- it is made up of decomposers, which are heterotrophic organisms, mainly fungi, and bacteria

Detritus → Earthworm → sparrow → Falcon

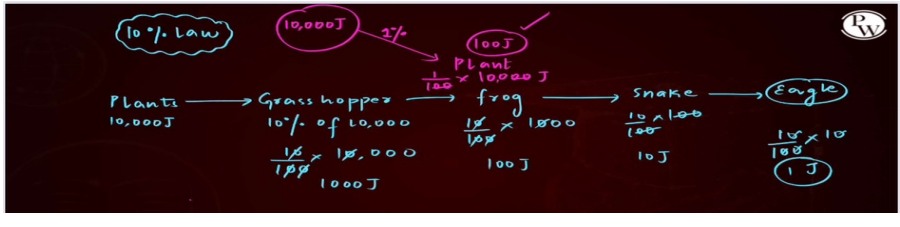
- in terrestrial ecosystem, a much larger fraction of energy flows through detritus food chain, than through grazing food chain

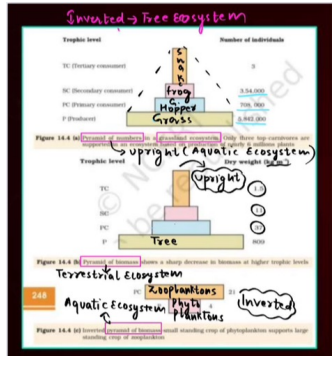
Parasitic food chain

- size of the organisms finally reduces with higher trophic levels (parasite)
- Example- Tree → Herbivore birds → lice & bugs

Ecological pyramids

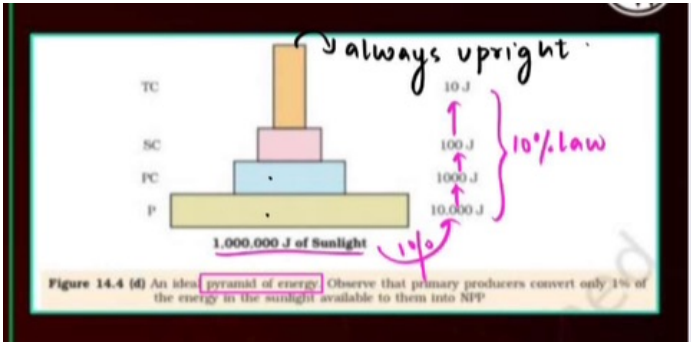
- graphical representation of various ecological parameters at the successive trophic levels
- Common parameters:
- 1) number of individual
- 2) biomass
- 3) energy at different trophic levels





Limitations of ecological pyramid

- it does not take into account the same species belonging to two or more trophic levels
- it, assume a simple food chain, that never exist in nature. It does not accommodate a food web
- Saprophytes are not given any place in ecological pyramids, even though they play a vital role in the ecosystem



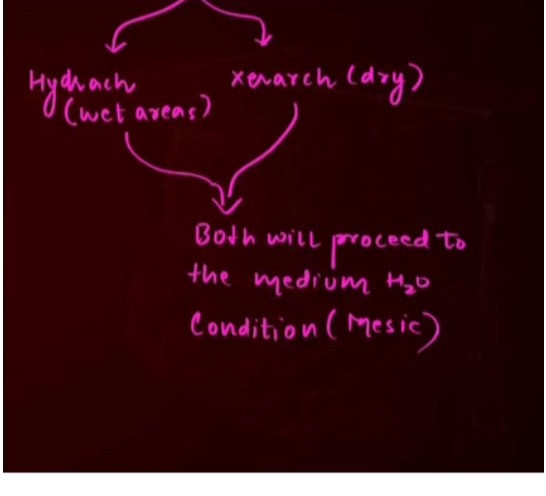
Ecological succession (by hult)

- gradual and fairly predictable changes in the species composition of a given area

Types of successional communities

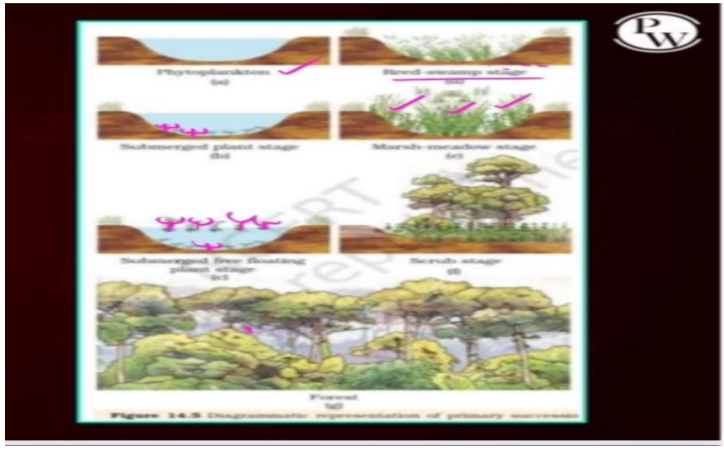
- pioneer community
- 1)first biotic community- example- lichens on rock, phytoplankton and zooplankton in Pond
- 2) transitional or seral community
- 3) climax community- last community, stable, eg - forest

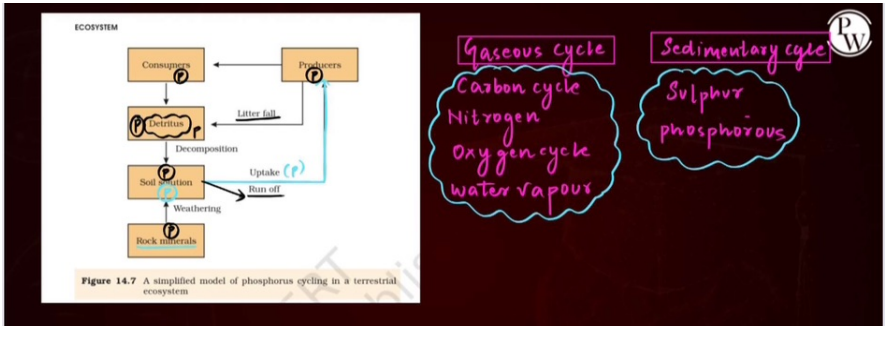
Types of succession



Ecosystem characteristics that change during succession

- change in diversity of species, some species colonise an area and their population become more numerous , population of other species, decline and even disappear
- little diversity to high degree of diversity, increase in the number of species
- Total biomass increases
- Increase in humus content soil
- aquatic or dry conditions to mesic conditions I.e, both hydrarch and xerarch succession lead to medium water condition, neither to dry nor too wet
- Vegetation changes in turn, affect food and shelter for various types of animals.Thus, as succession proceeds the number and types of animals and decomposers also change





Standing crop

- each trophic level has a certain mass of living material at a particular time called as the standing crop
- Standing crop is measured as the mass of living organism or the number in a unit area
- Measurement of biomass in terms of dry weight is more accurate

Standing state

- organisms need a constant supply of nutrients to grow reproduce and regulate various body functions
- The amount of nutrient such as carbon, nitrogen, phosphorus, calcium, etc. Present in the soil at any given time is called standing state.
- it varies in different kinds of ecosystem and also on a seasonal basis